



Probability Generating Functions

• Mark Scheme

Further Statistics 1 Chapter 7: Probability Generating Functions. M = method mark A = accuracy mark (depends on M)
B = independent mark ft = follow through

Section A – Probability generating functions

- 1 $G_X(t) = 0.1t + 0.6t^2 + 0.3t^3$: powers of t B1; coefficients B1.
- 2 $G_X(1) = 9k = 1 \Rightarrow k = \frac{1}{9}$ B1; $\frac{1}{9}(4 + 4t + t^2)$, so $P(X = 1) = \frac{4}{9}$ B1.
- 3 $G_X(t) = \sum_{x=0}^{\infty} t^x \frac{e^{-\lambda} \lambda^x}{x!}$ M1; $= e^{-\lambda} \sum \frac{(\lambda t)^x}{x!} = e^{-\lambda} e^{\lambda t}$ M1; $= e^{\lambda(t-1)}$ A1.
- 4 $G_X(t) = \sum_{x=1}^{\infty} t^x q^{x-1} p = pt \sum_{x=1}^{\infty} (qt)^{x-1}$ M1; geometric series with first term pt and ratio qt M1; $= \frac{pt}{1-qt}$ A1; valid for $|qt| < 1$, i.e. $|t| < \frac{1}{q}$ B1.
- 5(a) $G_X(t) = \frac{t^2}{3} (1 - \frac{2}{3}t)^{-1} = \frac{t^2}{3} (1 + \frac{2}{3}t + \dots)$ M1; $P(X = 2) = \frac{1}{3}$, $P(X = 3) = \frac{2}{9}$ A1.
- 5(b) $\frac{t^2}{3} \sum_{r=0}^{\infty} (\frac{2}{3}t)^r$ M1; coefficient of t^x : $P(X = x) = \frac{1}{3} (\frac{2}{3})^{x-2}$ A1; for $x = 2, 3, 4, \dots$ A1.
- 5(c) $|\frac{2}{3}t| < 1$ M1; $|t| < \frac{3}{2}$ A1.
- 5(d) $P(\text{even}) = \frac{1}{2} [G_X(1) + G_X(-1)] = \frac{1}{2} (1 + \frac{1}{5})$ M1; $= \frac{3}{5}$ A1.

Section B – Mean and variance

- 6 $G'_X(t) = 0.3 + t$ M1; $E(X) = G'_X(1) = 1.3$ A1.
- 7 $G'_X(t) = np(q + pt)^{n-1}$ M1 A1; $G'_X(1) = np(q + p)^{n-1} = np$, since $q + p = 1$ A1.
- 8 $G'_X(t) = \frac{2}{9}(1 + 2t)^2$, so $E(X) = 2$ B1; $G''_X(t) = \frac{8}{9}(1 + 2t)$, $G''_X(1) = \frac{8}{3}$ M1; $\text{Var}(X) = \frac{8}{3} + 2 - 4 = \frac{2}{3}$ A1.
- 9 $G'_X(t) = \lambda e^{\lambda(t-1)}$ M1; $E(X) = G'_X(1) = \lambda$ A1; $G''_X(1) = \lambda^2$ M1; $\text{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$ A1.
- 10(a) $\frac{t+t^2}{4} (1 - \frac{1}{2}t)^{-1} = \frac{t+t^2}{4} (1 + \frac{1}{2}t + \dots)$ M1; $P(X = 1) = \frac{1}{4}$, $P(X = 2) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$ A1.
- 10(b) $G'_X(t) = \frac{(1+2t)(4-2t) + 2(t+t^2)}{(4-2t)^2}$ M1; $= \frac{4+8t-2t^2}{(4-2t)^2}$ A1; $G'_X(1) = \frac{10}{4} = \frac{5}{2}$ A1.
- 10(c) $G''_X(t) = \frac{(8-4t)(4-2t) + 4(4+8t-2t^2)}{(4-2t)^3}$ M1 A1; $G''_X(1) = \frac{8+40}{8} = 6$ A1; $\text{Var}(X) = 6 + \frac{5}{2} - \frac{25}{4} = \frac{9}{4}$ A1.

Section C – Sums and uniqueness

- 11 $G_{X+Y}(t) = e^{2(t-1)} \times e^{3(t-1)} = e^{5(t-1)}$ M1; this is the PGF of $\text{Po}(5)$, so by uniqueness $X + Y \sim \text{Po}(5)$ A1.
- 12 $G_Y(t) = [G_X(t)]^n$ M1; $= (0.3 + 0.7t)^n$ A1; so $Y \sim B(n, 0.7)$ B1.
- 13 $G_Y(t) = t^2 G_X(t^3) = t^2 e^{4(t^3-1)}$ B1; $G'_Y(t) = 2t e^{4(t^3-1)} + 12t^4 e^{4(t^3-1)}$ M1; $E(Y) = 2 + 12 = 14$ A1.
- 14 $G_{X+Y}(t) = \frac{1}{9}(1+2t)(1+t+t^2)$ M1; $= \frac{1}{9}(1+3t+3t^2+2t^3)$ A1; $P(X+Y=2) = \frac{3}{9} = \frac{1}{3}$ A1; $E(X+Y) = \frac{1}{9}(3+6+6) = \frac{5}{3}$ B1.
- 15(a) $G_Y(t) = [G_X(t)]^3$ M1; $= \left(\frac{0.4t}{1-0.6t} \right)^3$ A1.



15(b) Negative binomial, $r = 3$, $p = 0.4$ *B1*; by uniqueness, since this is the PGF of that distribution (the number of trials needed for 3 successes) *B1*.

15(c) $G'_Y(t) = 3[G_X(t)]^2 G'_X(t)$, with $G'_X(t) = \frac{0.4}{(1-0.6t)^2}$ *M1 A1*; $E(Y) = 3 \times 1 \times \frac{0.4}{0.16} = 7.5$ *A1*.

15(d) $G_Y(t) = 0.064t^3(1-0.6t)^{-3} = 0.064t^3(1+1.8t+\dots)$ *M1*; $P(Y=4) = 0.064 \times 1.8 = 0.1152$ *A1*.

Roar-level challenge

16 $G_X(t) = \frac{1}{6}(t+t^2+\dots+t^6) = \frac{t(1-t^6)}{6(1-t)}$; total has PGF $\frac{t^3(1-t^6)^3}{216(1-t)^3}$ *M1*; need the coefficient of t^7 in $(1-3t^6+\dots)(1-t)^{-3}$ *M1*; $(1-t)^{-3} = \sum \binom{r+2}{2} t^r$ *B1*; coefficient $= \binom{9}{2} - 3\binom{3}{2} = 36 - 9 = 27$ *M1*; probability $\frac{27}{216} = \frac{1}{8}$ *A1*.